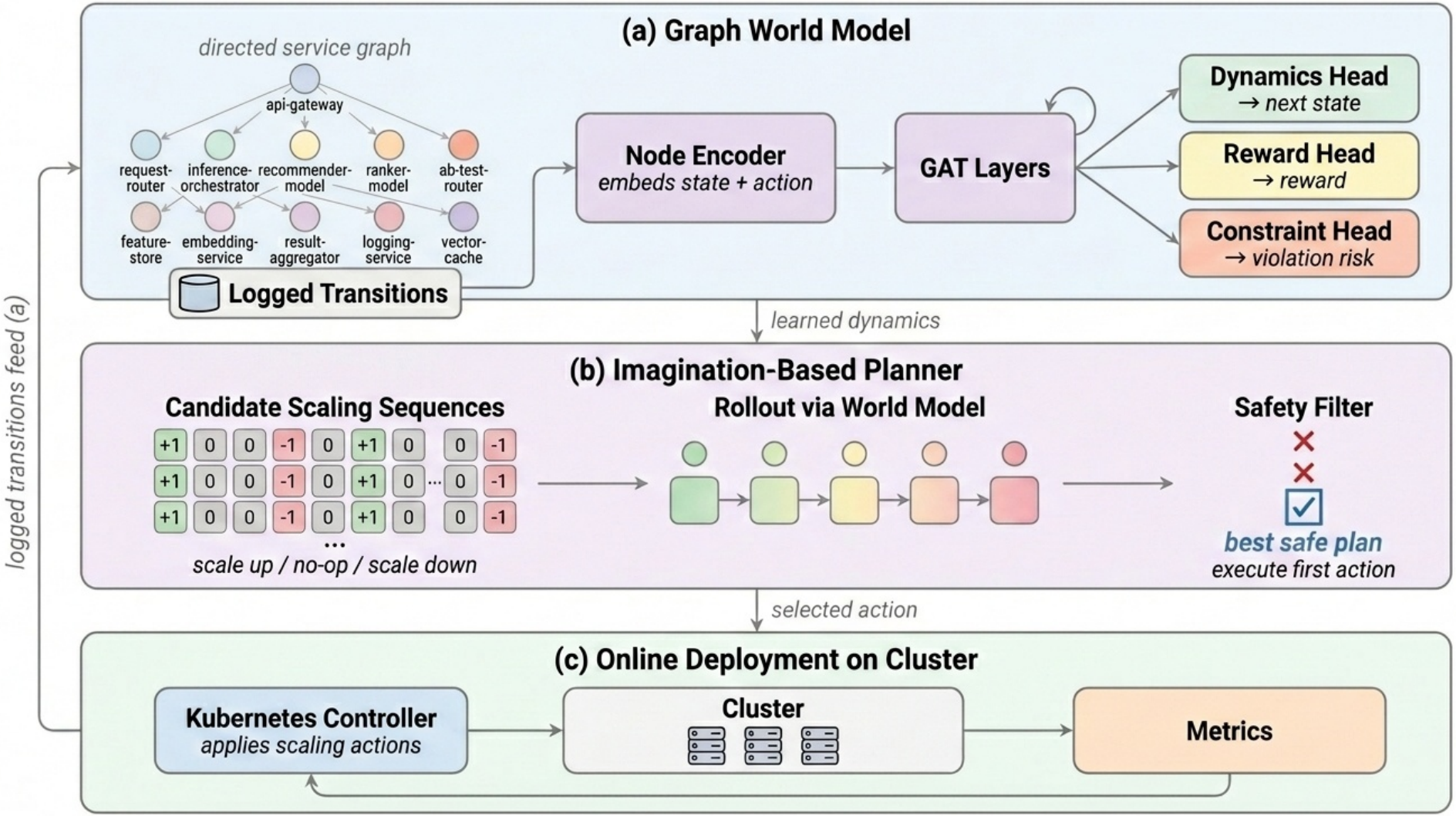
(a) Graph World Model
directed service graph
api-gateway
request-router
inference-orchestrator
recommender-model
ranker-model
ab-test-router
feature-store
embedding-service
result-aggregator
logging-service
vector-cache
Logged Transitions
Node Encoder
embeds state + action
GAT Layers
Dynamics Head
→ next state
Reward Head
→ reward
Constraint Head
→ violation risk
learned dynamics
(b) Imagination-Based Planner
Candidate Scaling Sequences
+1 0 0 -1 0 +1 0 0 -1
+1 0 0 -1 0 +1 0 ... 0 -1
+1 0 0 -1 0 +1 0 0 -1
...
scale up / no-op / scale down
Rollout via World Model
Safety Filter
best safe plan
execute first action
selected action
(c) Online Deployment on Cluster
Kubernetes Controller
applies scaling actions
Cluster
Metrics
logged transitions feed (a)